

Module 5: Advanced Pattern Mining & Applications

Multilevel Rules, Antimonotonic Constraints, Colossal & Closed Patterns

Module V: Advanced Pattern Mining, Classification & Clustering — Complete Tracker

Topic 37: Pattern Mining Road Map (Itemsets to Multidimensional)	Topic 38: Multilevel Pattern Mining (Concept Hierarchy Abstractions)
Topic 39: Multidimensional Pattern Mining (Multiple Predicates)	Topic 40: Constraint-Based Pattern Mining (Pushing Constraints)
Topic 41: Mining High-Dimensional Data & Colossal Patterns	Topic 42: Mining Compressed Patterns (Closed vs. Maximal Itemsets)
Topic 43: Mining Approximate Patterns & Error Tolerances	Topic 44 & 45: Pattern Exploration & Real-World Domain Applications
PYQ Block A: Classification (Decision Trees, Naïve Bayes, Backprop & Metrics)	PYQ Block B: Clustering (K-Means, PAM Medoids, DBSCAN & BIRCH CF-Trees)

Topic 37 – 40: Advanced Pattern Mining Methodologies

Advanced Mining Paradigm	Algorithmic Mechanism & Description	Key Benefit / Example
1. Multilevel Association	Mines rules across different concept hierarchy levels (e.g., `Milk => Bread` vs. `Dairy => Baked Goods`).	Uncovers granular patterns at lower levels and generalized trends at higher levels.
2. Multidimensional Association	Involves multiple predicates or dimensions (e.g., `age(X, "20..29") ^ income(X, "50K..60K") => buys(X, "Laptop")`).	Discovers rich customer demographic profiles beyond simple single-item co-occurrences.
3. Constraint-Based Mining	Pushes user-defined constraints (Antimonotonic, Monotonic, Succinct) directly into the mining algorithm.	Massively prunes the candidate search space; produces only actionable, relevant rules.
4. Closed vs. Maximal Patterns	• Closed Itemset: An itemset X is closed if no proper superset has the same support. • Maximal Itemset: An itemset X is maximal if no proper superset is frequent.	Compresses millions of frequent itemsets into compact loss-less or lossy representations.

★ PYQ-CRITICAL BLOCK A: Classification Algorithms [UPLOADED PYQ]

1. Naïve Bayesian Classification [UPLOADED PYQ – IT426 End]:

$$P(C_i | \mathbf{X}) = \frac{P(\mathbf{X} | C_i)P(C_i)}{P(\mathbf{X})} = \frac{P(C_i) \prod_{k=1}^n P(x_k | C_i)}{P(\mathbf{X})}$$

Class-Conditional Independence Assumption: Assumes that the effect of an attribute value on a given class is completely independent of the values of other attributes.

2. Backpropagation Neural Network Learning [UPLOADED PYQ – IT335 End]:

$$w_{ij} \leftarrow w_{ij} + \Delta w_{ij} = w_{ij} + (l)\text{Err}_j O_i$$

- For output layer neuron k : $\text{Err}_k = O_k(1 - O_k)(T_k - O_k)$ - For hidden layer neuron j : $\text{Err}_j = O_j(1 - O_j) \sum_k \text{Err}_k w_{jk}$

3. Classification Evaluation Metrics [UPLOADED PYQ]:

Metric	Formula	Practical Interpretation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Overall proportion of correct predictions.
Precision	$\frac{TP}{TP+FP}$	Out of all predicted positives, how many are truly positive?
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	Out of all actual positive cases, how many were discovered?
Error Rate	$\frac{FP+FN}{TP+TN+FP+FN} = 1 - \text{Accuracy}$	Proportion of misclassified instances.

★ PYQ–CRITICAL BLOCK B: Clustering Algorithms [UPLOADED PYQ]

Algorithm [UPLOADED PYQ]	Clustering Paradigm	Algorithmic Principle	Key Advantages & Tradeoffs
1. K-Means [UPLOADED PYQ]	Partitioning (Centroid-based)	Iteratively assigns objects to nearest mean centroid μ_j and recalculates centroids to minimize $SSE = \sum \sum \ x - \mu_j\ ^2$.	Fast $O(tkn)$; sensitive to noise and outliers; requires k upfront.
2. PAM (K-Medoids) [UPLOADED PYQ]	Partitioning (Medoid-based)	Uses actual representative data points (medoids) instead of mean averages. Iteratively evaluates quality swaps.	Highly robust to noise and outliers; computationally expensive $O(k(n - k)^2)$.
3. DBSCAN [UPLOADED PYQ]	Density-Based	Grows clusters from high-density core points ($\geq \text{MinPts}$ within ϵ -radius). Marks sparse points as noise.	Discovers arbitrary-shaped clusters; immune to noise; doesn't require k upfront.
4. BIRCH [UPLOADED PYQ]	Hierarchical (Feature Tree)	Builds an in-memory CF-Tree (Clustering Feature Tree) where each node contains $CF = (N, \text{LS}, SS)$.	Linearly scalable $O(n)$ for massive datasets with single scan of disk.

🧠 M5 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ — IT335 End] Explain DBSCAN clustering algorithm with core points, border points, and noise points. (8 Marks)

DBSCAN relies on two parameters: radius ϵ and density threshold **MinPts**:

- Core Point:** A point p is a core point if $|N_\epsilon(p)| \geq \text{MinPts}$ (at least **MinPts** points lie within distance ϵ).
- Border Point:** A point q is not a core point, but falls within the ϵ -neighborhood of some core point p .
- Noise Point:** Any point that is neither a core point nor a border point.

Clustering: Connects all density-reachable core points into clusters and assigns border points to their adjacent core clusters.